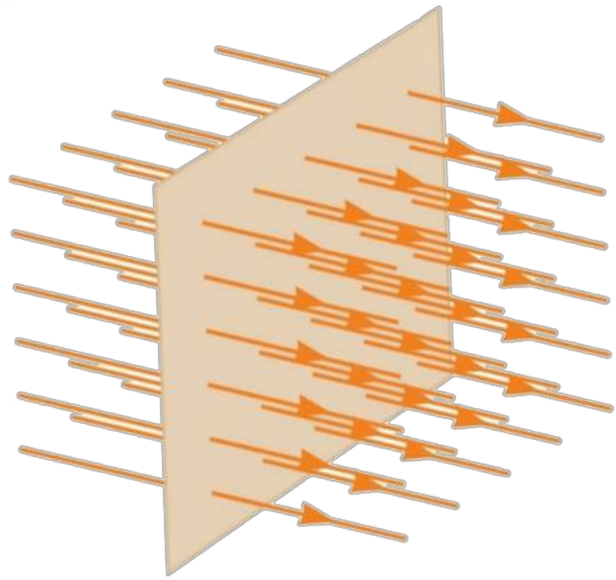
Chapter (24) Gauss's Law:-

24.1 Electric Flux:-



Electric Flux:- is the number of electric field lines that Pass a plane of area (A) Perpendicular to the field.

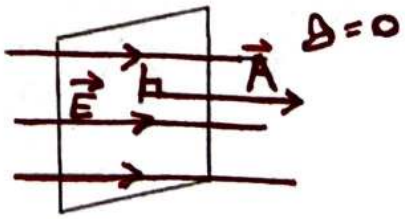
$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

$$\Phi_E = [N/C] \cdot [m^2] \equiv N \cdot m^2/C$$

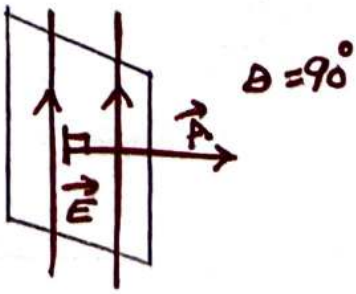
$\vec{A}$:- area vector that is has magnitude of area and direction perpendicular to the surface.

θ :- is the angle between $\vec{E}$ and $\vec{A}$.

$$\Phi_{max} = EA\cos(0) = EA$$



$$\Phi_{min} = EA\cos(90) = \text{zero}$$



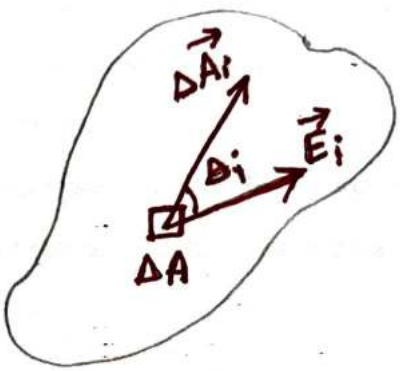
For nonuniform surface, we divide the surface into small equal areas each one = ΔA

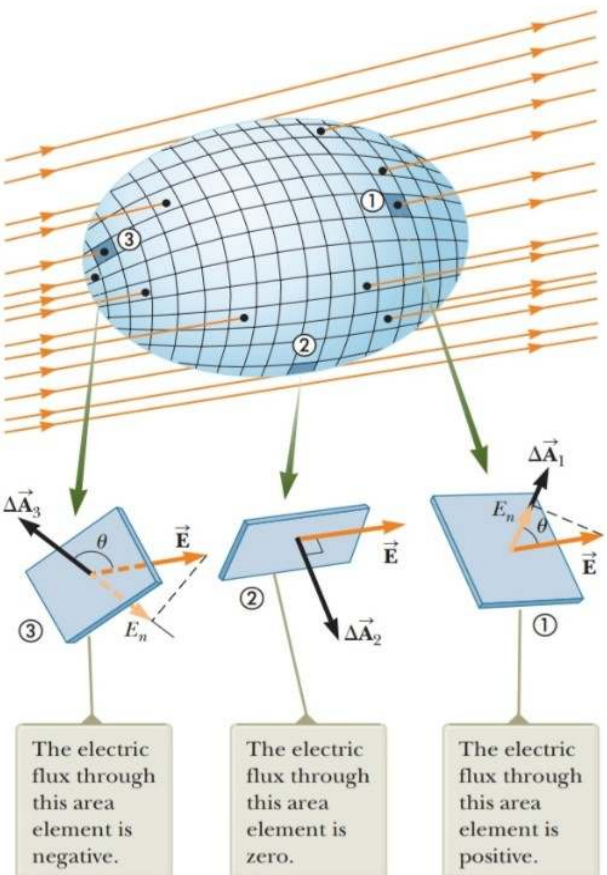
$$\Delta \Phi_E = E_i \, \Delta A_i \cos\theta_i$$

$$\Phi_E = \lim_{\Delta A_i \to 0} \sum \vec{E}_i \cdot \Delta \vec{A}_i$$

$$= \int_{\text{surface}} \vec{E} \cdot d\vec{A}$$

$$= \oint \vec{E} \cdot d\vec{A} \quad , \text{ for closed surface.}$$





Quick Quiz 24.1 Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere? (a) The flux and field both increase. (b) The flux and field both decrease. (c) The flux increases, and the field decreases. (d) The flux decreases, and the field increases. (e) The flux remains the same, and the field increases. (f) The flux decreases, and the field remains the same.

the answer is (e), the flux will remain the same as the same number of electric field lines will pass through the surface. but $\vec{E}$ will inereasc; $\boxed{E = \frac{Kq}{r^2}}$, when the radius of the sphere is halved E will be double twice.

$$E' = \frac{Kq}{(\frac{1}{2}r)^2} = 4\frac{Kq}{r^2} = 4E$$

$$\Phi = EA = \frac{Kq}{r^2} \cdot 4\pi r^2$$

$$\Phi' = E'A' = \cancel{4}\frac{Kq}{r^2} \cdot 4\pi\frac{r^2}{\cancel{4}} = \frac{Kq}{r^2} \cdot 4\pi r^2$$

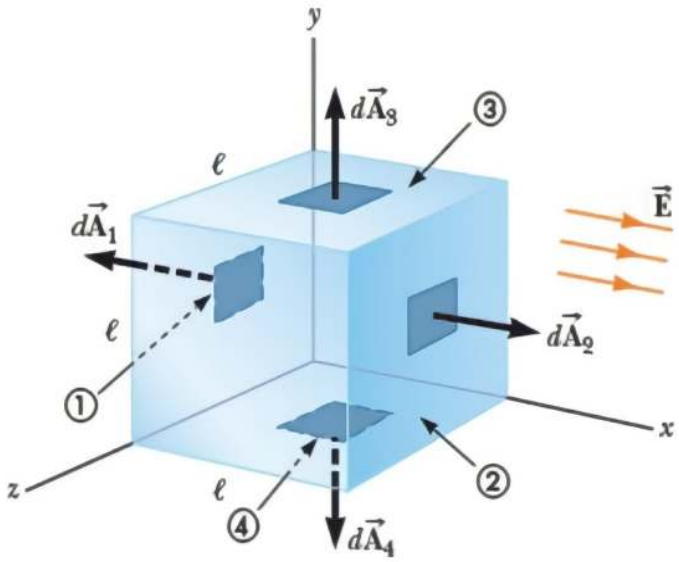
$$\therefore \boxed{\Phi = \Phi'}$$

Example (24.1) :- Flux through a cube :-

$$\Phi_E = \int_1 \vec{E}\cdot d\vec{A} + \int_2 \vec{E}\cdot d\vec{A}$$

$$+ \int_3 \vec{E}\cdot d\vec{A} + \int_4 \vec{E}\cdot d\vec{A}$$

$$+ \int_5 \vec{E}\cdot d\vec{A} + \int_6 \vec{E}\cdot d\vec{A}$$



$$\Phi_E = \int_1 E\cos(180)\,dA + \int_2 E\cos(0)\,dA + \int_3 E\cos(90)\,dA$$

$$+ \int_4 E\cos(90)\,dA + \int_5 E\cos(90)\,dA + \int_6 E\cos(90)\,dA$$

$$= -EA + EA + 0 + 0 + 0 + 0$$

$$= 0$$

24.2 Gauss's Law :-

the net charge flux through any closed surface (gaussian surface) is equal to the net charge inside the surface divided by ϵ_0.

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

Consider +ve point charge in the center of sphere of radius r.

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$\Phi = \oint E \, dA \cos(0) = EA$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \cdot 4\pi r^2$$

$$\boxed{\Phi = \frac{Q}{\epsilon_0}}$$

Notes:-

1- the net flux through any closed surface surrounding a point charge Q is $\frac{Q}{\epsilon_0}$.

2- Φ is independent of the shape of the surface.

3- Φ is independent of the radius (r).

4- Φ through closed surface without charge inside it equal zero.

Conceptual Example 24.2: Flux Due to a Point Charge

A spherical gaussian surface surrounds a point charge *q*. Describe what happens to the total flux through the surface if (A) the charge is tripled, (B) the radius of the sphere is doubled, (C) the surface is changed to a cube, and (D) the charge is moved to another location inside the surface.

A) $\Phi = \frac{Q_{in}}{\epsilon_0}$ $\qquad Q'_{in} = 3Q_{in}$

$$\Phi' = \frac{Q'_{in}}{\epsilon_0} = 3\frac{Q_{in}}{\epsilon_0} = 3\Phi$$

B) Φ is constant; beause it is not depend on(r).

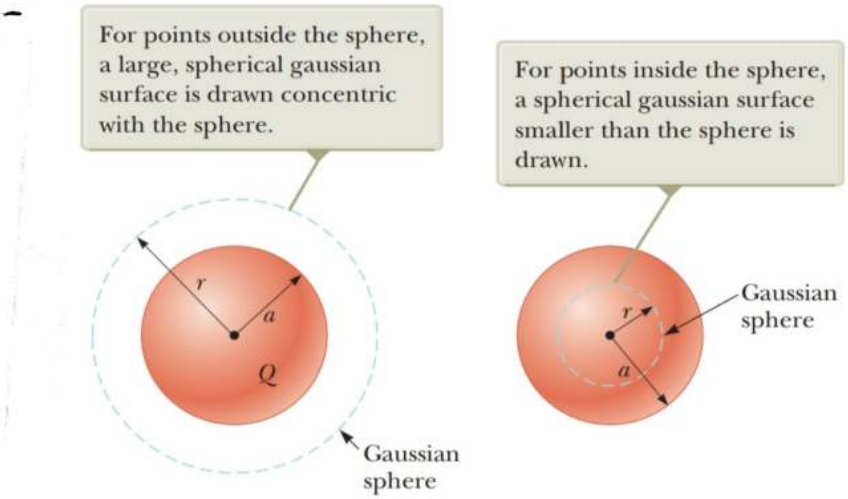
C) Φ is constant because it's independ on the shape.

D) Φ is constant because gauss'sLaw refer to the total charge Q inside the surface.

24.3 Application of Gauss's Law to various charge Distributions:-

Gauss's Law is useful for determining E, when the charge distribution is symmetric such as sphere, cylinder and plane sheet.

Ex(24.3) - a spherically symmetric charge distribution:-



A) E outside the sphere

$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

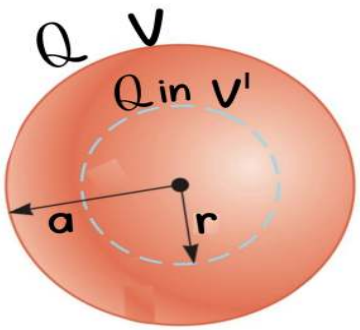
$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$E = K \frac{Q}{r^2}$ the same for point charge.

(for $r > a$).

B) $\vec{E}$ inside the sphere.



$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$\rho = \frac{Q_{in}}{V'} = \frac{Q}{V}$$

$Q_{in} = \rho V'$, where V' ... of gauss's Law.

$$Q_{in} = \rho V' = \rho \frac{4}{3}\pi r^3$$

$$\oint \vec{E} \cdot d\vec{A} = E \cos(0) \oint dA = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$E = \frac{\rho r}{3\epsilon_0} \text{ , but } \rho = \frac{Q}{\frac{4}{3}\pi a^3}$$

$$E = \frac{Q}{\frac{4}{3}\pi a^3} \cdot \frac{r}{3\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^3} r$$

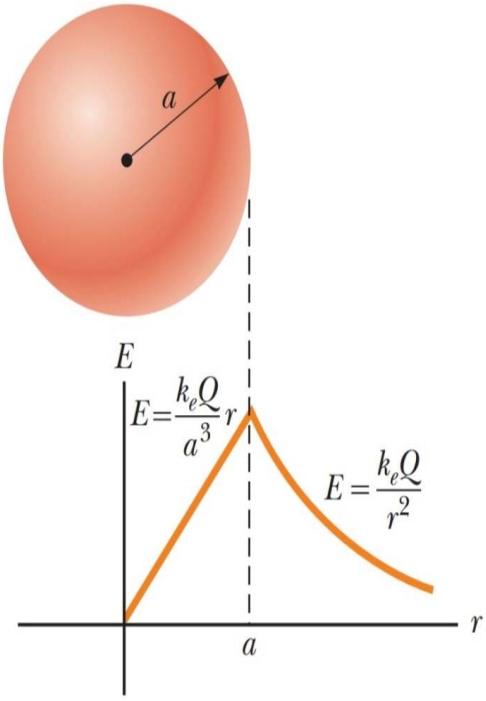
$$E = \frac{K Q r}{a^3} \quad (\text{for } r < a)$$

at $r = 0 \Rightarrow E = 0$

What if $r = a$??

$$E_{outside} = \lim_{r \to a} \left(\frac{kQ}{r^2}\right) = \frac{kQ}{a^2}$$

$$E_{inside} = \lim_{r \to a} \frac{kQr}{a^3} = \frac{kQ}{a^2}$$



Example:- for spherical shell find $\vec{E}$ inside and outside the shell?

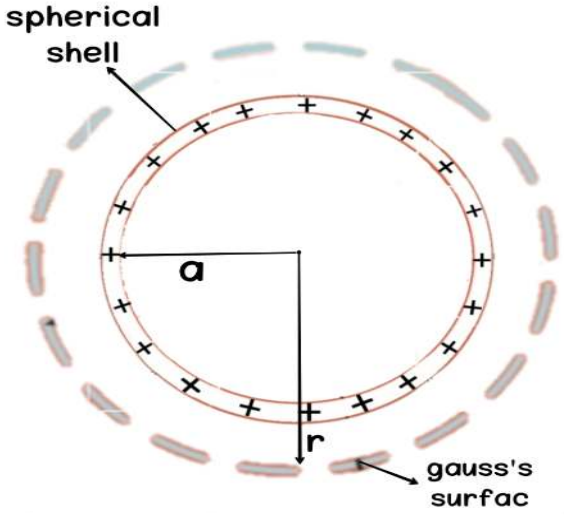
(a) outside the shell

$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

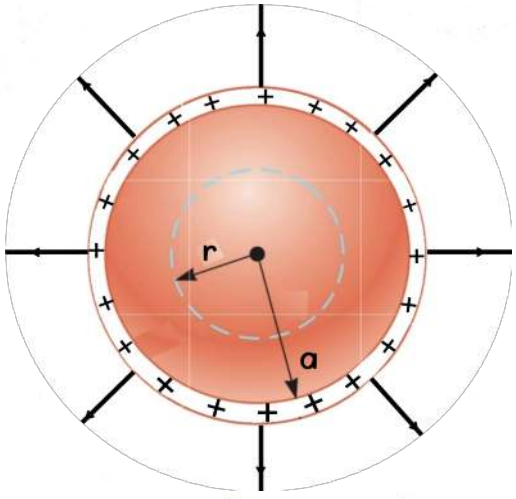
$$E = \frac{kQ}{r^2}, \text{ for, } r > a$$



(b) inside the shell

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = 0$$

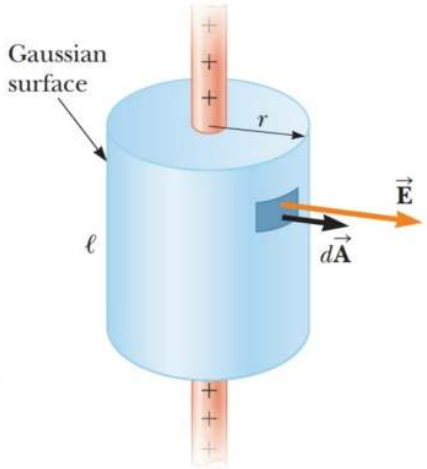


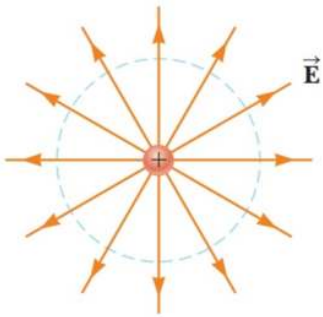
$E = 0$, inside the spherical shell where Q = zero

Ex(24.4):- A cylindrically symmetric charge distribution:-

flux through upper and lower surfaces = zero;

$\theta = 90^\circ$.





$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$E\cos(0) \oint dA = \frac{Q_{in}}{\epsilon_0}$$

$$E\,(2\pi r \ell) = \frac{\lambda \ell}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \cdot \frac{2}{2} = \frac{2\lambda}{4\pi\epsilon_0 r}$$

$$E = \frac{2K\lambda}{r}, \quad E \propto \frac{1}{r}$$

Ex(24.5): A Plane of charge:-

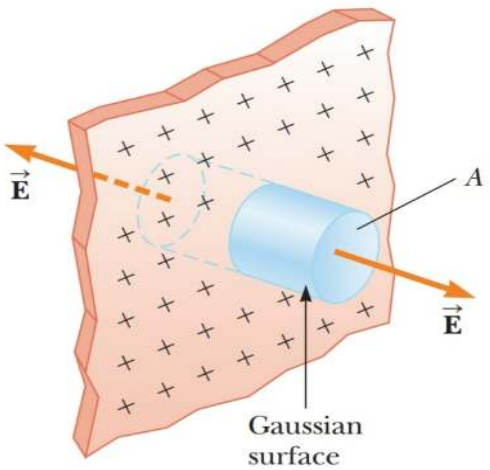
$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$

$$= EA + EA + 0$$

$$= 2EA$$

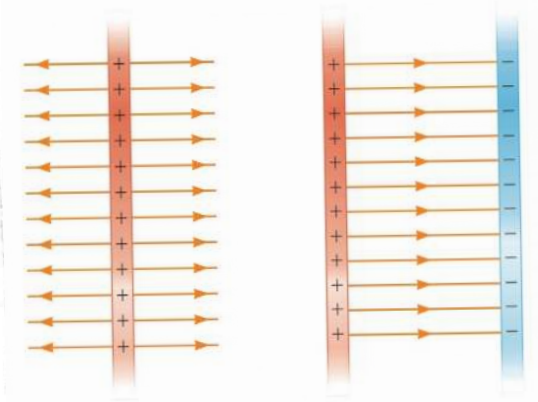
$$\Phi = \frac{Q_{in}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} = 2EA$$

$$E = \frac{\sigma}{2\epsilon_0}, \text{ uniform } \vec{E}$$



what if we have two parallel plate with +ve and -ve charge

$\vec{E}$ between the plates is uniform and zero elsewhere.



$$EA = \frac{Q_{in}}{\epsilon_0}$$

$$EA = \frac{\sigma A}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{\epsilon_0}} \equiv \text{uniform Electric field.}$$

24.4 Conductors in electrostatic equalibrium :-

when there is no motion of charge with in the conductor, the conductor is in electrostatic equalibrium.

Properties of a conductor of electrostatic equalibrium :-

1- if an isolated conductor carries a charge the charge resides on the surface.

2- E is zero every where inside the conductor

wherer the conductor is solid or hollow.

3- the electric field just outside a charged conductor is perpendicular to the surface of conductor and has a magnitude of $\frac{\sigma}{\epsilon_0}$.

4- On an irregularity shaped conductor, the surface charge density is greatest at locations where the radius of curvature of the surface is smallest.

$\sigma = \frac{Q}{A}$, $A \propto r^2$

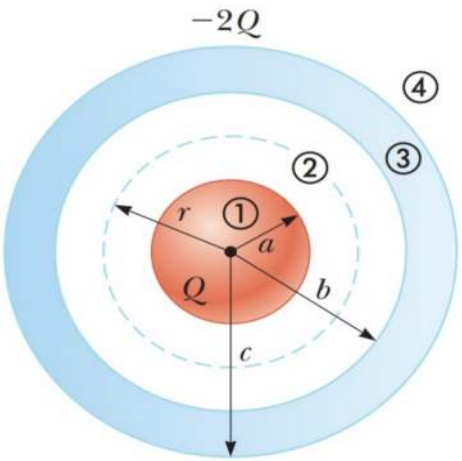
Smaller (r) → Smaller (A) → larger (σ)

Quick Quiz 24.3 Your younger brother likes to rub his feet on the carpet and then touch you to give you a shock. While you are trying to escape the shock treatment, you discover a hollow metal cylinder in your basement, large enough to climb inside. In which of the following cases will you not be shocked? (a) You climb inside the cylinder, making contact with the inner surface, and your charged brother touches the outer metal surface. (b) Your charged brother is inside touching the inner metal surface and you are outside, touching the outer metal surface. (c) Both of you are outside the cylinder, touching its outer metal surface but not touching each other directly

answer is (a).

Ex (24.7):- A sphere inside a spherical shell:-

in inner sphere is insulating sphere with radius ⓐ. with charge Q the spherical shell of 2-radii ⓐ and ⓑ and $-2Q$ charge.



$$E_1 = \frac{K Q r}{a^3} \quad (\text{for } r < a)$$

$$E_2 = K \frac{Q}{r^2} \quad (\text{for } a < r < b)$$

$$E_3 = 0 \quad ; Q_{in} = 0 \quad (\text{for } b < r < c)$$

$$E_4 = -\frac{KQ}{r^2} \quad (\text{for } r > c)$$

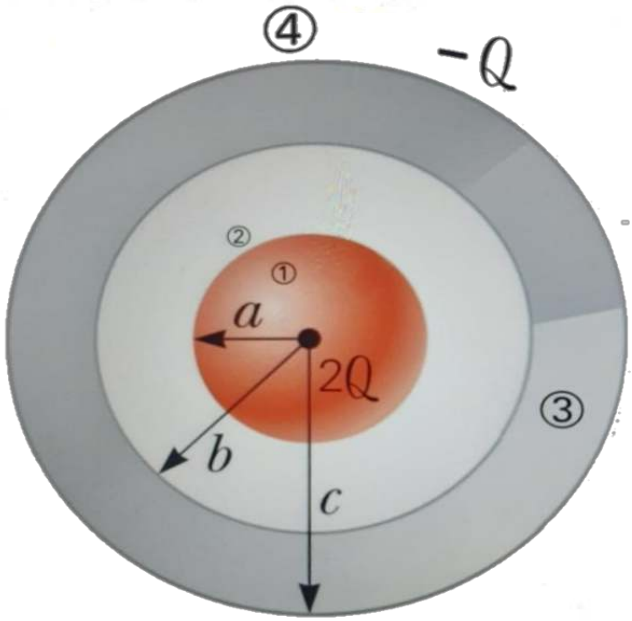
$$Q_{in} = -2Q + Q$$

$$= -Q$$

what if the inner sphere is conducting sphere ??

$E_1 = 0$; $Q_{in} = 0$.

Example:- conducting sphere of radius (a) and charge (2Q) surrounding by conducting spherical shell of inner and outer radii (b) and (c) and charge -Q ↑ find E in the regions 1, 2, 3 and 4 ??



1- $E_1 = 0$; $Q_{in} = 0$

for $r < a$

2- $E_2 A = \frac{Q_{in}}{\epsilon_0}$

$E_2 (4\pi r^2) = \frac{2Q}{\epsilon_0}$

$E_2 = \frac{2KQ}{r^2}$ for $(a < r < b)$

3- $E_3 = 0$, $Q_{in} = 0$

4- $E_4 = \frac{KQ_{in}}{r^2} = \frac{K(2Q - Q)}{r^2} = \frac{KQ}{r^2}$